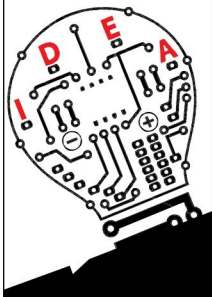


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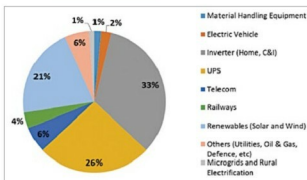


Fig. 3: Application-wise revenue share of lead-acid battery market in 2020E (Source: IESA)

Batteries in demand

Electrochemical storage technology, primarily driven by secondary batteries (lead-acid and lithium-ion), captures a major share of the Indian energy storage market. Lead-acid batteries can be segmented into flooded batteries (used in deep-cycle applications) and VRLA batteries (suitable for standby power with low depth-of-discharge characteristics). Advances in related technology to increase battery life-cycle, charge acceptance, efficiency and discharge performance, as well as low upfront costs will influence the growth of lead-acid battery segment.

IESA forecasts the lead-acid battery market size to reach around US\$ 4 billion by 2020, representing CAGR of 13 per cent for the period 2015-20 (Fig. 3). Currently, inverter and UPS applications take a major share (60 per cent) of the stationary and motive battery market. In coming years, home inverter market growth will stagnate at 7-8 per cent due to improved power scenario and use of common power backup in new townships and societies of urban areas.

Increased automation in C&I segment with scattered locations of IT offices and data centres in Tier 2 and Tier 3 cities makes use of UPS mandatory in India. This will lead to a healthy growth (10-15 per cent) in this segment.

Renewable integration, especially solar PV and wind, as well as the government's thrust on electric vehicles (EV) and e-rickshaws already makes this market very attractive.

Telecom is another promising segment for battery use, but newer

technologies like Li-ion and flow batteries are giving tough competition to use of lead-acid batteries for telecom towers. Expansion of the railway network, ambitious target of metro trains and introduction of bullet trains make railways an attractive segment with growth prospect of 4

per cent for lead-acid battery manufacturers.

In India, Li-ion batteries are emerging as an environment-friendly, rechargeable power source in comparison to conventional lead-acid, nickel-cadmium and nickel-metal-hydrate batteries. These are increasingly being used in consumer electronics products (such as smartphones and tablets) and healthcare equipment (like surgical devices and respirators). With continuing technological developments, Li-ion batteries are also being used in the automotive industry, particularly in electric vehicles. However, in the absence of proper domestic manufacturing, Indian manufacturers need to import Li-ion batteries from other Asian countries like China, Taiwan and South Korea.

According to a TechSci report, the Li-ion batteries market in India will witness double-digit growth at over 35 per cent annually till 2020. The government's efforts to minimise carbon emissions and toxicity level in the environment are likely to drive the adoption of Li-ion batteries across major end-use industries including consumer electronics and automotive. In addition, the government has announced tax exemption on electric vehicles powered by Li-ion batteries, which is anticipated to drive the market further. Owing to their compact size, lightweight and high durability, Li-ion batteries are expected to witness wider adoption in telecom and data communication applications as well as energy storage systems. **EFY**